Correcting lineshape distortions using ShimNet

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ShimNet is a neural network for post-acquisition correction of NMR spectra distorted by magnetic field inhomogeneity. The network is trained using synthetic spectra convolved with “natural” distortions measured in a calibration procedure. The software is described in the paper:

The hands-on will cover:

1. The use of ShimNet program and web service.
2. The collection of calibration spectra and the extraction of distortions
3. The generation of training spectra
4. The example neural network training
5. The comparison of networks obtained using different training parameters.

The ShimNet is a standalone Python script, generally run from the command line. However, for better understanding and consistent visualization of the results, we will work today in the Google Colab environment.

To start, log in to Google Colab and type:

```
from google.colab import drive
drive.mount('/content/drive')
%cd /content
!git clone -b workshops https://github.com/center4ml/shimnet.git
!mv shimnet /content/drive/MyDrive/shimnet_workshop
```

This will connect Colab to Google Drive and download the ShimNet package with example data from GitHub.

Open the notebook named ShimNet_workshop1 from your Google Drive (folder Workshop_Colab_Notebooks).

Keep Google Drive open in a separate tab or window.

Finally, run the first cell to enter the shimnet folder and install all the required Python packages and download the data.

1. The use of ShimNet program and webservice.

The cell named *ShimNet Predict* contains the command starting the ShimNet reconstruction. The following arguments are provided to the program:

- input spectrum (text file with two columns)
- model (trained network)
- config file

Run the cell to correct the distorted spectrum using ShimNet.
Run the next cell named** Plot results **(set the proper filename).
The input argument can be changed to run ShimNet on the other spectra (listed below). Config and model are the same for all listed spectra, as all were acquired at the same 600 MHz.

`"Asarone_deshimmed"`
Spectrum of 14 mM Asarone in MeOD (intentionally) deshimmed using Z1 and Z2 shims

`"Asarone_reduce_volume"`
Sample shimmed, then removed from the magnet, 20 ul subtracted, sample inserted again and spectrum re-axcquired without shimming

`"lineshape_deshimmed"`
Spectrum of 1% CHCl3 in acetone-d (intentionally) deshimmed using Z1 and Z2 shims

`"multi_singlets_deshimmed"`
Spectrum of a mixture of …….. giving several singlets in a 1H NMR spectrum.

In the next cell named *High intensity range* we test the program on the spectrum with large H2O peak. Such high dynamic range is rarely present in the training data and thus the network performs a bit worse.

Run the cell named** High intensity range **and examine the resulting spectra.
Which result looks better?

Two spectra are used:

`"Cresol_Red_after_different_sample"`
Spectrum of 0.78 mM Cresol Red in (wet) DMSO with huge H2O peak present. Shims adjusted using different sample.The range of signal intensities is very high, rarely present in the training data. Thus, the correction is suboptimal. Compare with the next spectrum.

`"Cresol_Red_after_different_sample_cut"`
Spectrum of 0.78 mM Cresol Red in (wet) DMSO with huge H2O peak zeroed. Shims adjusted using different sample.

Open the web browser and go to
https://huggingface.co/spaces/NMR-CeNT-UW/ShimNet
Upload selected dataset and run the processing.

2. The collection of calibration spectra and the extraction of distortions

ShimNet requires the collection of calibration spectra acquired inhomogeneous magnetic field of various spatial profiles generated by missetting the shim coil currents in a controlled way.

Open the script sweep_XXX from the calibration_loop folder on Google Drive to see how the data is collected on the Agilent machine.

Open the Colab notebook named ShimNet_workshop2

Run the first cell to connect to Google Drive and install packages

The second cell demonstrates the concept of convolution. Run it to see how the simulated spectrum is distorted by a function causing Gaussian broadening. All peaks are distorted in the same way.

In the third cell we try to deconvolve the distortion function (Gaussian) from the distorted spectrum. Change the noise level () and the deconvolution window () and observe the effect.

In the fourth cell, we will extract the experimental shim coil response functions (SCRFs). Run it to see the plots of example CHCl3 peaks and deconvolved distortion functions. Deconvolved functions are stored in XXXX and will be used in the next step to generate training spectra. The extraction script uses the following parameters:

`# input`
`data_dir` - folder with raw FIDs from the calibration loop
`opti_fid_path = "./opti_spectra/opti_total6.fid"` - folder with raw FID of optimally shimmed calibration sample

`# output`
`spectra_file = "./sample_run/total.npy"` - file with parts of spectra used for deconvolution
`spectra_file_names = "./sample_run/total.csv"` - file with names of fids from calibration
`opi_spectrum_file = "./sample_run/opti.npy"` - file with part of opti spectrum used for calibration
`responses_file = "./sample_run/scrf_shimator9.pt"` - extracted distortion functions
`losses_file = "./sample_run/losses_scrf_shimator9.pt"` - file with loss values of respective distortion functions deconvolution

3. The generation of training spectra

ShimNet is trained using thousands or millions of simulated spectra generated using:

- experimental distortion functions extracted from the calibration spectra (above)
- multiplets patterns from the text file (the format of Ambit NMR software) corresponding to typical small molecules
- parameters from the config.yaml file

Normally, the generation of training data and actual network training are realized within one script. For the demonstration purposes we split them here.

Open the notebook named ShimNet_workshop3.

Run the first cell to mount Google Drive and install required packages.
Run the second cell to download the file with multiplet patterns.
Run the third cell to download the file with 10,000 multiplet patterns (Enamine database** https://enamine.net/compound-collections/real-compounds/real-database**).
Run the fourth cell to generate the example spectrum 1H.
Examine the generated spectra. Play with parameters e.g.

4. The example neural network training

Open the notebook named XXXXXX.

Run the first cell to

5. The comparison of networks obtained using different training parameters.